



Punc'data
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Punc'data

Tutorial for assignment

Purpose of this tutorial

- This tutorial is given to help you navigate through data importation in Punc'data and assignment of high resolution mass spectrometry data
- Calibration and the use of canvas (to make figures) will not be addressed here
- The data used in this tutorial is available in the source code of Punc'data (<https://github.com/WTVoe/puncdata>)

Folder : Examples\testdata_cellulosepyrolysis_POSITIVE_raw.csv

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1. Method

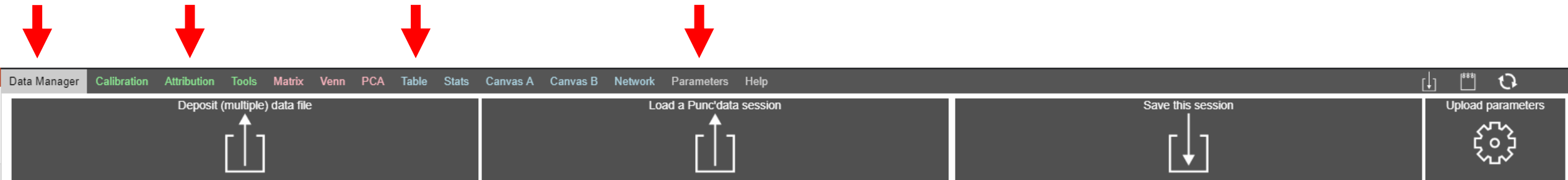
The assignment method in Punc'data includes both a systematic « De Novo » approach and a mass difference network approach. The former is the default one, while the latter is adapted to complex mixtures (DOM, NOM, oils, polymers...)

A few interesting studies on the subject:

- Kunenkov et al. Anal. Chem. 2009 <https://doi.org/10.1021/ac901476u>
- Tziotis et al. Eur. J. Mass Spectrom. 2011 <https://doi.org/10.1255/ejms.1135>
- Kilgour et al. Anal. Chem. 2012 <https://doi.org/10.1021/ac301339d>

2.1. Opening Punc'data

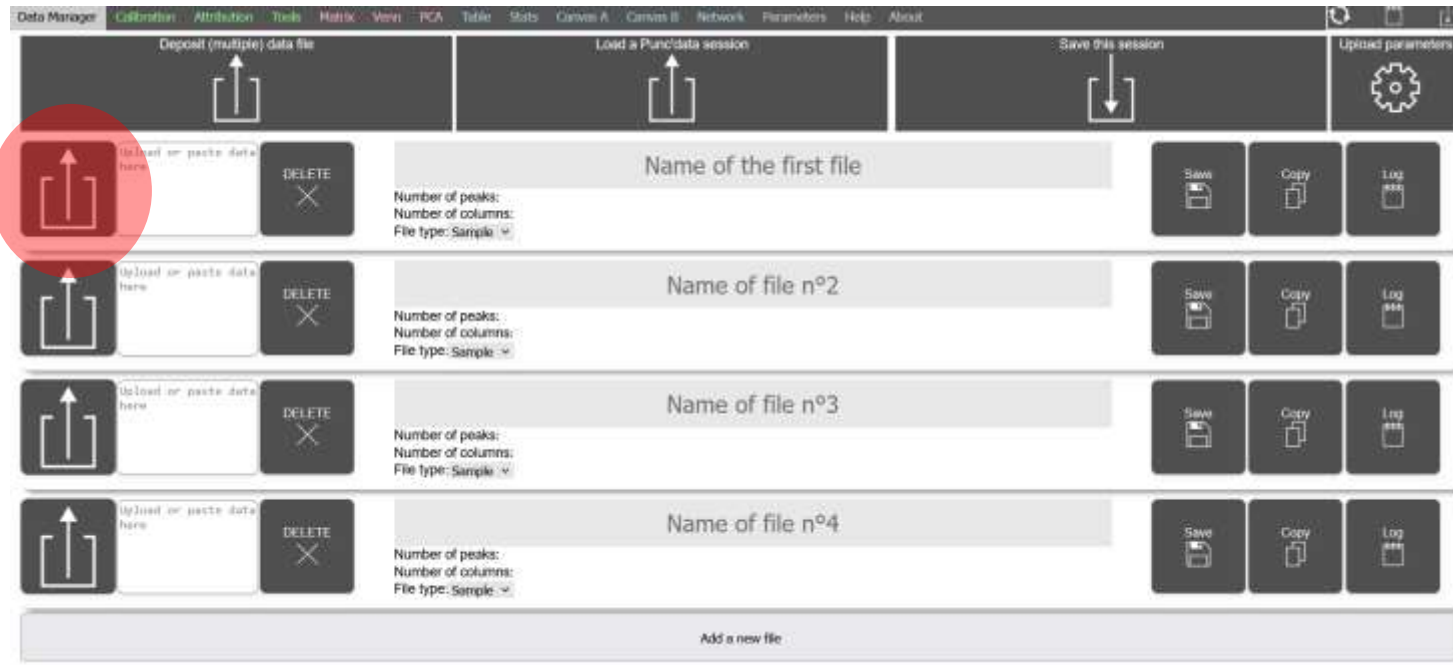
- Start Punc'data, with a downloaded version or through wtvoe.github.io/puncdata/
- Tabs at the top help you navigate through the functions of Punc'data
- The ones you will need for this tutorial are highlighted below:



2.2 Importing data

- Download the test data on Github :
[Examples\testdata_cellulosepyrolysis_POSITIVE_raw.csv](#)

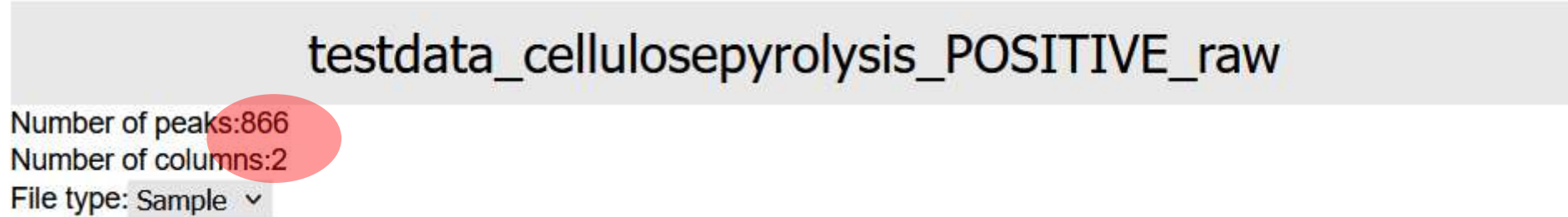
Punc'data only accepts peaklists, not profile spectra. Lines should be for peaks, columns for data related to each peak
Any column order can be accepted.



- Press the button highlighted in red and import your file
- *Alternatively you can paste it from a spreadsheet (CTRL+V) into the text area next to the highlighted button*
- You can then rename your file

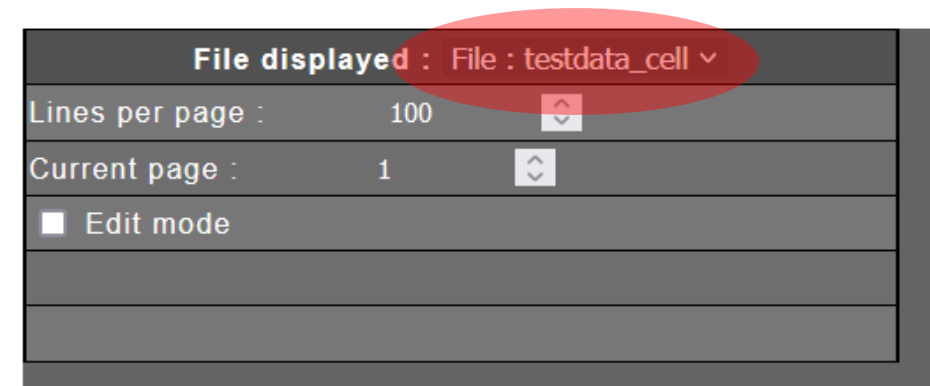
2.3 Verifying that the import is correct

- A number of peaks and a number of columns should appear once the import is complete



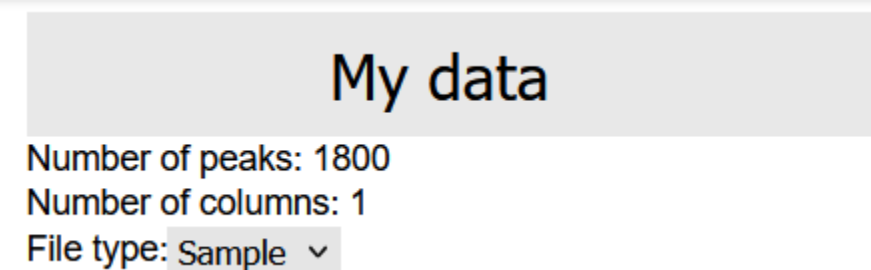
- Go to the tab « TABLE » and select your file in « file displayed » :
- Your file should appear below

For the test file: the columns are « m/z » and « I », with a line for each m/z value

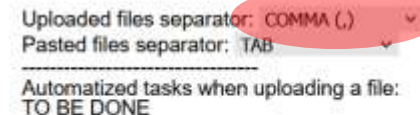


2.3 Common issue

- A common issue when importing data is related to the separator between columns in the uploaded file. When a problem occurs, on tab « DATA MANAGER » the number of columns will display only 1



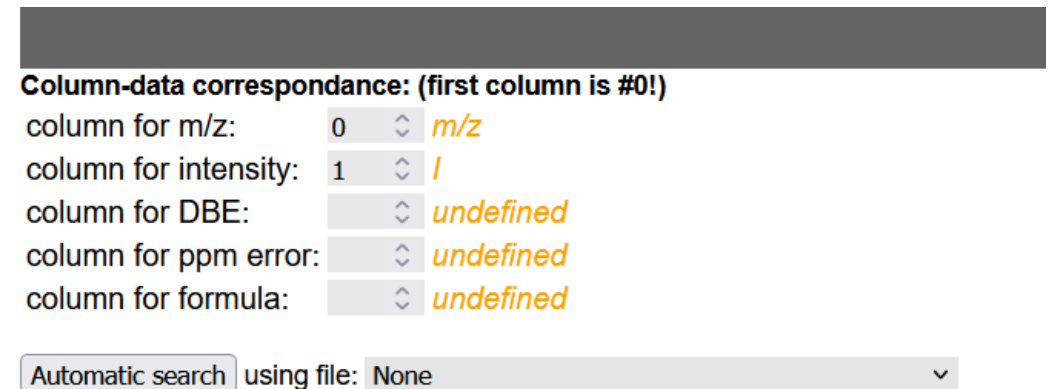
- To fix this, press the button « upload parameters » on the top-right, with the gear icon
- A menu opens, select the correct option for your file in « uploaded file separator » and press save
- Press « DELETE » for your file and re-upload it.



2.3 Verifying that the import is correct

- Go to the tab « PARAMETERS »
- Punc'data should automatically identify your m/z and I columns, on the leftmost menu:
- If this is not correct for your data, you can set them by hand. Beware, the first column is numbered « 0 »

TIP: by selecting a file below (after « automatic search ») and pressing the button, the columns names in orange will be updated to show the correct names



Column-data correspondance: (first column is #0!)

column for m/z:	0	◇	m/z
column for intensity:	1	◇	I
column for DBE:		◇	undefined
column for ppm error:		◇	undefined
column for formula:		◇	undefined

Automatic search using file: None ▾

3.1 Assignnement: the basics

- Go to the third tab : « ATTRIBUTION »

Set up your
parameters
here

The screenshot shows a software interface with a dark grey header bar and a large white content area. The header bar contains several sections: 'File to attribute : None' with a dropdown arrow; a list of checkboxes for 'Isotopy' (checked), 'Seeds' (checked), 'Network' (unchecked), 'Passes' (checked), 'Filtering' (checked), 'Posttreatment', and 'Log results'; a 'Remove isotopes from list' section with a 'Choose elements' button and an 'Isotope tagging tolerance (mDa)' input field set to '0,1'; a 'Charts' section with a 'Pie Chart' and 'Opacity : 100 %' setting; and a 'Main chart cfg menu' section with 'Chart 1', 'Chart 2', 'Chart 3', and 'Data' options. On the far right of the header bar are three buttons: 'Compute' (with a calculator icon), 'Validate results' (with a checkmark icon), and 'Import config' (with a document icon). Below the header bar, the main content area is mostly empty, with the word 'Hello' visible in the top left corner. A large red bracket on the left side of the header bar points to the 'Isotopy' and 'Seeds' checkboxes, with the text 'Set up your parameters here'. A red bracket on the right side of the header bar points to the 'Compute', 'Validate results', and 'Import config' buttons, with the text 'Start the process and save the results'. A red text label 'Your results will be displayed here' is centered in the main content area.

Assignment parameters

Display parameters

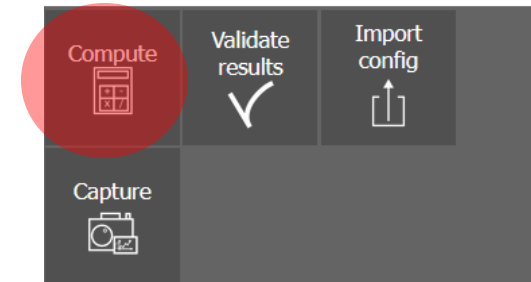
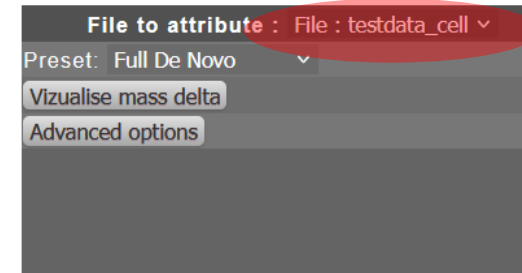
Start the process and save the results

Your results will be displayed here

3.1 Assignment: the basics

- Let's start with the leftmost menu. Select your file in « file to attribute »
- For now, leave the preset « Full De Novo ».
No mass differences will be computed
- Press « compute » on the rightmost menu

TIP: the menu « visualize mass delta » will be a useful option later to display a specific mass difference histogram for your chosen sample



3.1 Assignnement: the basics

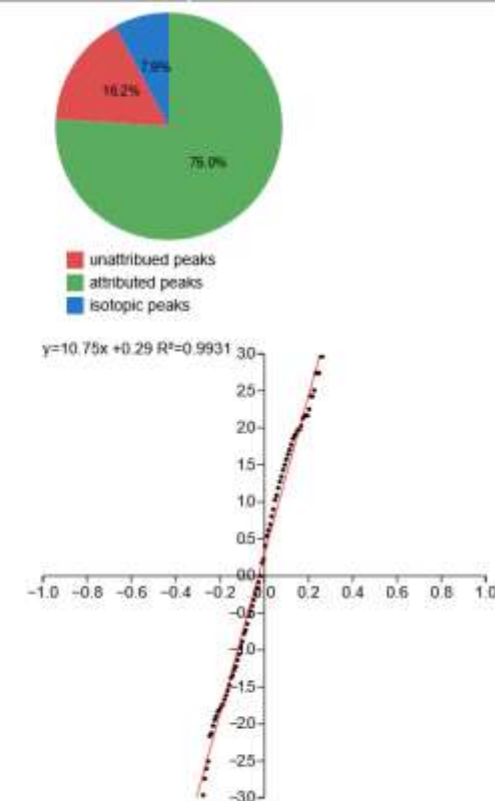
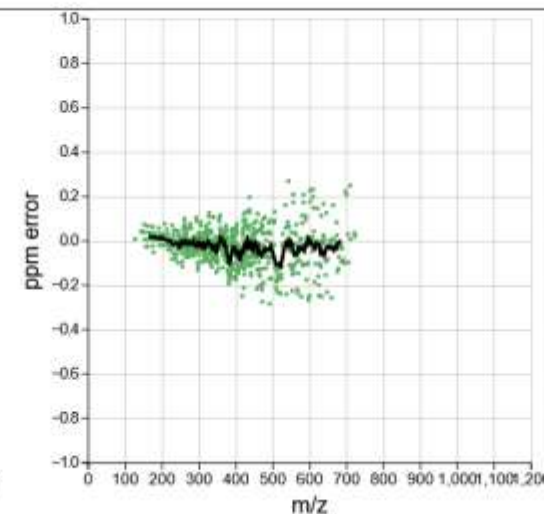
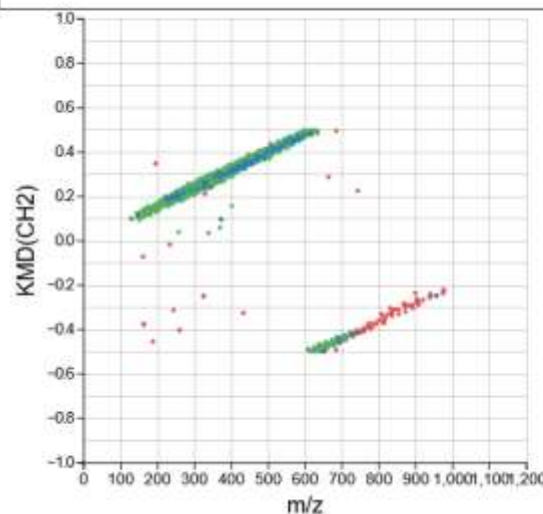
- You should get these results
- The text box displays the progress of the algorithm

If it ever happens that the last line isn't displayed, it means that the algorithm encountered a problem and stopped

```
isotope search finished. Found 68  
Seed search finished. Found 0  
Pass 1 build in 156ms  
Pass 1 attributed in 15ms, 657 attributions found  
Total Attributions found: 658 attributions found  
0.00% found by network  
Execution time: 199 ms  
Press "Validate results" if you want to keep results
```

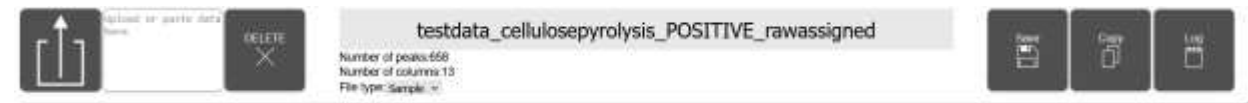
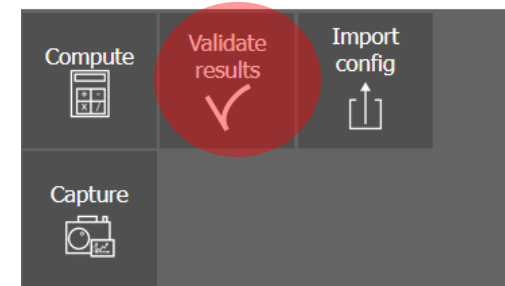
The pie chart represents the proportion of assigned peaks (with isotopes in blue) and non-assigned

These charts can be modified, this will be mentioned in the last part of this tutorial (part 5)

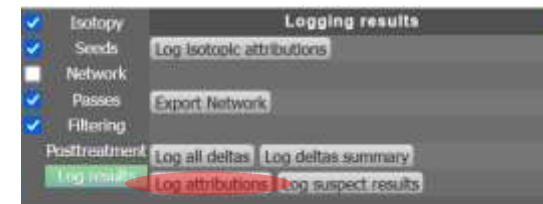


3.1 Assignment: the basics

- Once you are happy with your results, press « Validate results »
- You will be asked what to do with the results. For this example, let's select « ADD AS A NEW FILE ASSIGNED »
- Go back to the « DATA MANAGER » tab. Your results will appear as a new file, with 13 columns (*we will learn how to change this later*). You can then copy or save this dataset
- Alternatively, on the « ATTRIBUTION » tab, middle-left menu, you can press « Log results » and then « Log attributions ». The results table will be displayed

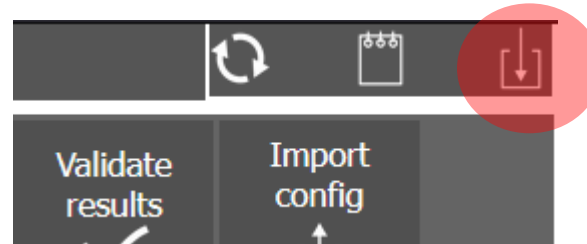
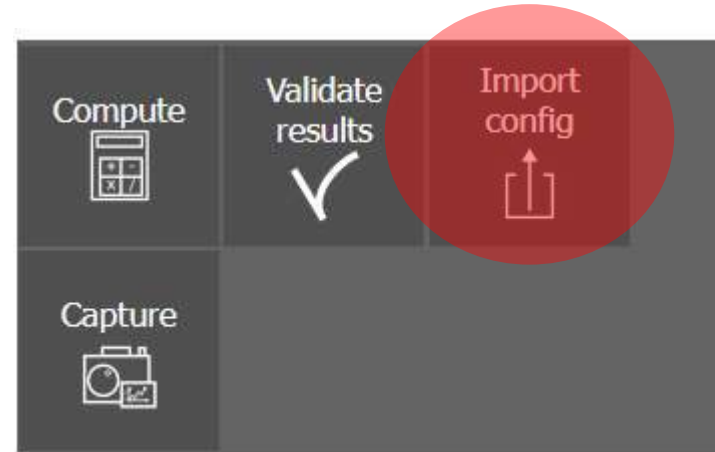


OR:



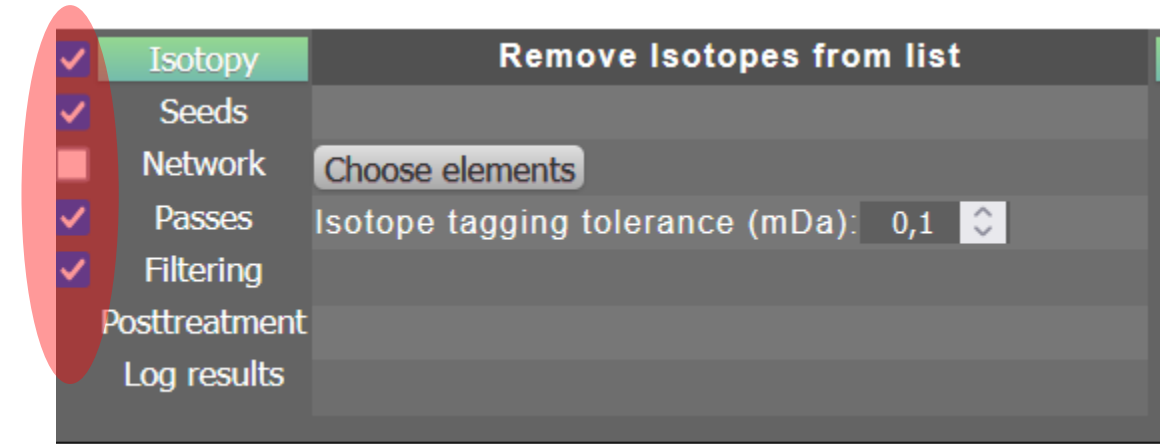
3.2 Import/export parameters

- If you already have customized parameters, you can import them by pressing « import config » and choosing your file (.json or .pdata)
- If you want to save your current parameters, press the top-right button next to the notebook. You can then export only the parameters (.json file) or the whole session, with datasets included (.pdata file)
- Alternatively, you can go to the « DATA MANAGER » tab to also save the session. The effect is identical.



3.3 Modify parameters

- The assignment process is segmented in subprocesses. They can be checked in/out as you wish on the middle-left menu. You can try to check in/out some of them and see the effects on the assignment results after pressing « compute » again.
- « Isotopy » tags the isotopic distributions
- « Seeds » assigns a list of predefined ionic formulae
- « Network » is for creating a mass difference network that will assist assignments
- « Passes » is used to modify the De Novo assignment procedure
- « Filtering » is used to defined rules to filter out unrealistic formulae



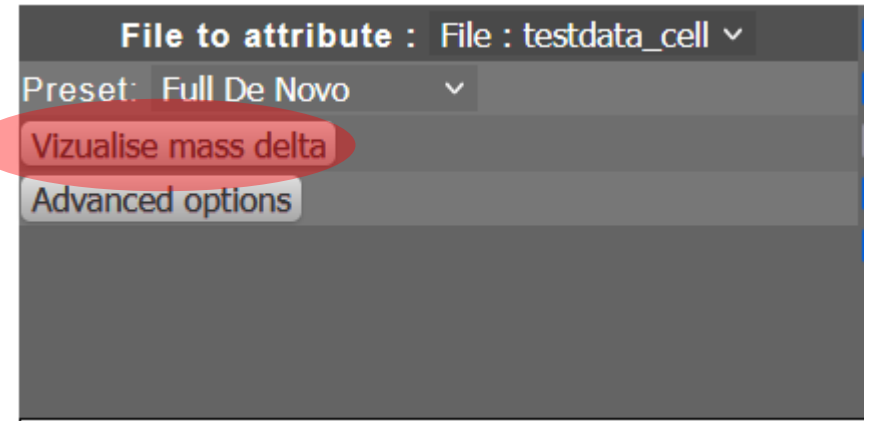
Kind et al. BMC Bioinformatics 2007
<https://doi.org/10.1186/1471-2105-8-105>

3.3.1 Isotopy

- « Isotopy » searches for isotopic mass differences and tags the peaks. Press « choose elements » to select which one you need. *For the test sample, leave only ^{13}C .*

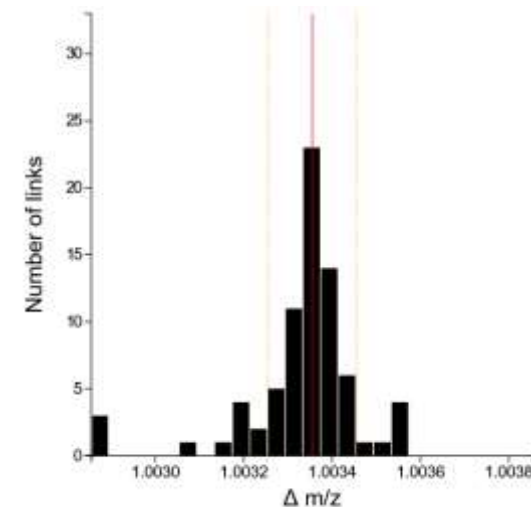
TIP: with « advanced edition » you can add other elements/isotopics distribution that could occur in your datasets and that you want to tag.

- The correct tagging tolerance depends on your sample. To estimate it, press « visualize mass delta » on the left menu and press « DRAW »
- Configure the width of the orange dotted lines with « Half-window tolerance »
- Once you find a value that encompasses your mass difference peak, use the same value for the isotope tagging tolerance. *Here we will use 0.2 mDa*



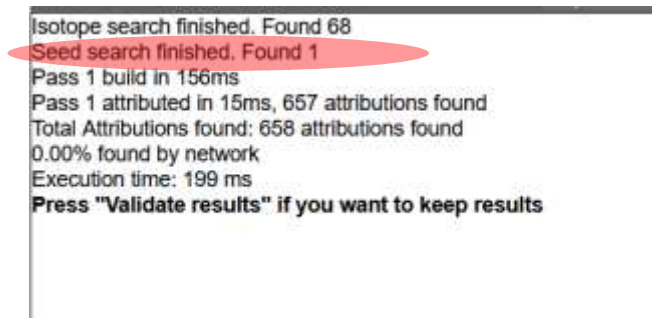
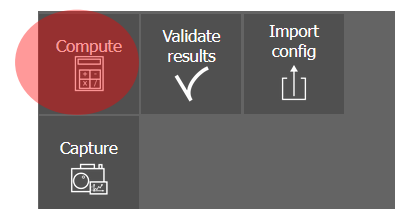
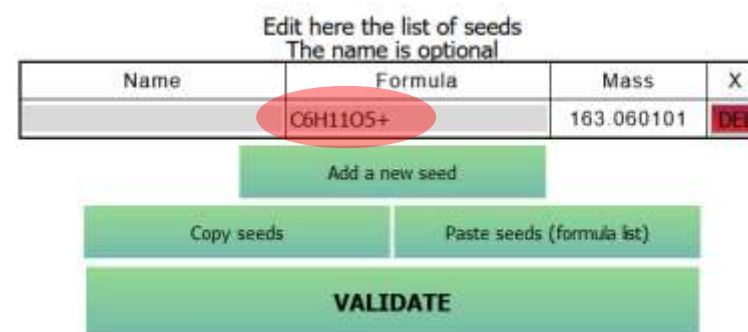
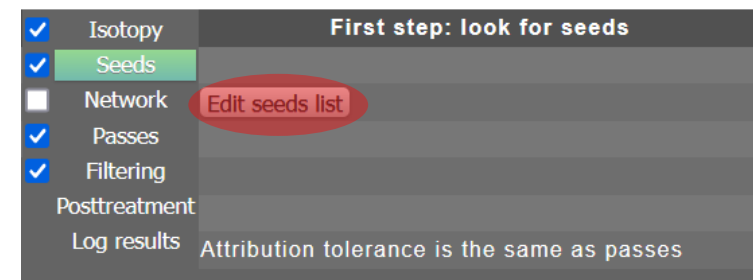
Vizualise the deltas around a specific $\Delta m/z$ value	
delta: isotope-13C	1,003355
Window width (mDa)	1
Number of bins	25
Half-window tolerance (mDa)	0,1

DRAW



3.3.2 Seeds

- You can add seeds, which will be the first formulae to be assigned. Press the button « Edit seeds list »
- You can paste a list of formulae from a spreadsheet (press paste seeds), OR you can write them one by one by pushing « Add a new seed »
- **WARNING:** you should use Ionic formulae, with a charge !
- For the example dataset, let's add **C6H11O5+** and press « VALIDATE »
- If you re-compute, the text box should now display: « seed search finished. Found 1 »

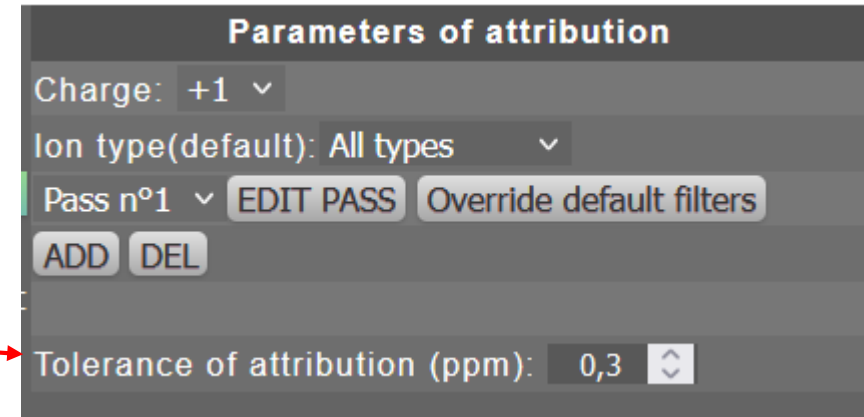


3.3.3 Networks

- The network submenu is a bit more complex. Go to **part 4** to better understand how to customize it best to your needs
- For now, leave the checkbox unchecked

3.3.4 Passes

- This menu helps you build (multiple) passes that will search the best element combination for every m/z value in your dataset within a **defined tolerance**
- For this sample, the ionization was done in APCI(+), so the correct charge is (+1) and the correct ion type is « All types ». *If you use ESI(-) for example, use charge (-1) and Ion Type « No radicals ».*
- *For this sample, you can leave the tolerance at 0.3 ppm*



Parameters of attribution

Charge: +1 ▾

Ion type(default): All types ▾

Pass n°1 ▾ EDIT PASS Override default filters

ADD DEL

Tolerance of attribution (ppm): 0,3 ▾

The algorithm:

- Pass 1 formulae are computed
 - The best fit is searched for every m/z value in your dataset
 - For a defined m/z value, If at least one formula is found within the tolerance of attribution (ppm) and respects the filtering rules, the peak is assigned. If there are multiple formulae, the one with the lowest error is chosen
 - If you used a network, it is explored with this new assignment as a seed
- The algorithm then proceeds to Pass 2 with the non-assigned peaks only

3.3.4 Passes

- Press « edit pass »
- A menu opens that allows you to customize the element list, the minimal and maximal value of each of them in any given formula
- Leave « mz Max » at -1 if you don't want to define a maximal m/z value computed.
- You can copy/paste other passes from a spreadsheet. You can also name your pass if needed
- When you press « VALIDATE » Punc'data alerts you if the pass exceeds 4 million possibilities, as computation times will get increasingly longer
- For this dataset, the default pass is already good enough since we will use the network later.



Edit the pass here

Pass Name:

mz Max: -1

Copy this pass Paste a pass

Element	min	max	DEL
C	0	40	DEL
H	0	60	DEL
O	0	20	DEL
N	0	1	DEL

Save and add a new element

VALIDATE

3.3.4 Passes

- The button « ADD » creates a new pass. It will be empty at creation. Select it, and press « EDIT PASS » to modify it
- For the test dataset, only one pass will be enough
- The button « DEL » deletes the currently selected pass
- The button « Override default filters » will open a menu that is a copy of the « Filtering » menu, but which can be modified **only for the currently selected pass**
- **If you want to try it, press « Toggle on overriding »**
- We won't need it for our current example, as we only have one pass

Parameters of attribution

Charge: +1 ▾

Ion type(default): All types ▾

Pass n°1 ▾ EDIT PASS Override default filters

ADD DEL

Tolerance of attribution (ppm): 0,3 ▴ ▾

☐ Toggle on overriding of golden rules for this pass only

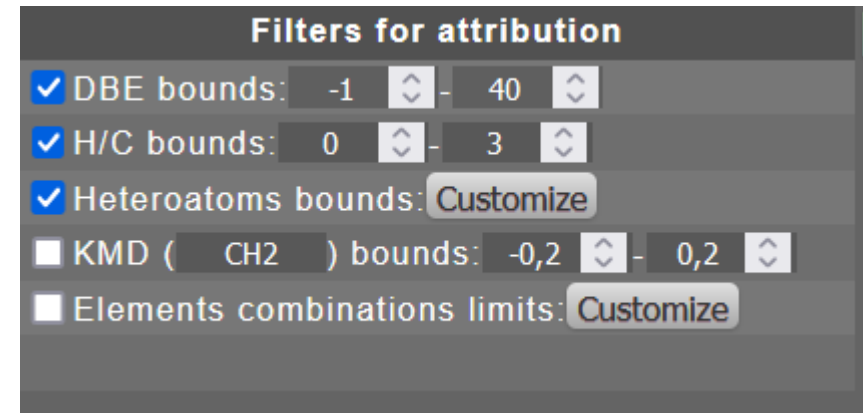
Ion type allowed : All types ▾

Toggle	Name	Bounds
☒	DBE	-1 ▴ ▾ 40 ▴ ▾
☒	H/C ratio	0 ▴ ▾ 3 ▴ ▾
☒	Hetero ratio	Customize
☐	KMD CH2	-0,2 ▴ ▾ 0,2 ▴ ▾
☐	Elements combinations	Customize

VALIDATE

3.3.5 Filtering

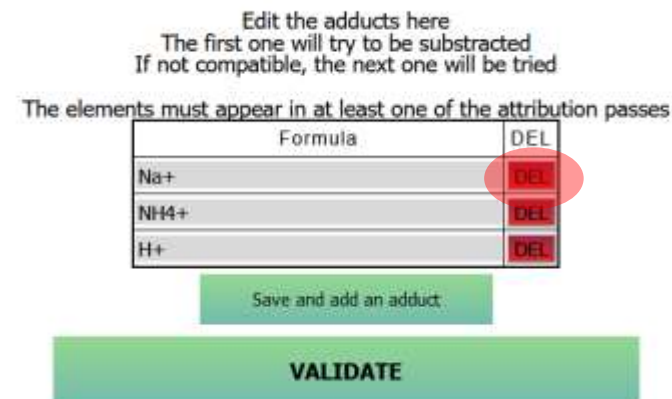
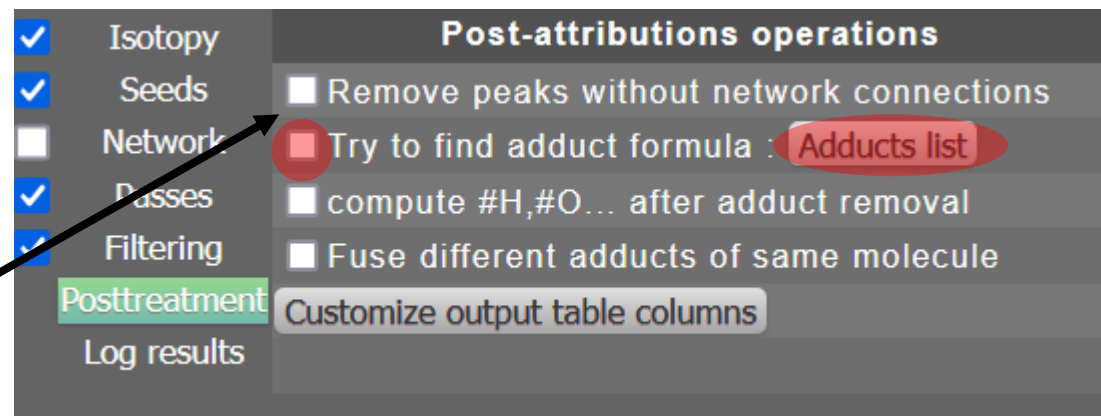
- The default parameters of this menu are inspired by the « Seven Golden Rules »
- These values are already good enough for a vast majority of datasets
- You can check in/out which rules are applied for every formula
- « KMD » (Kendrick mass defect) limits the assignment to only a subsection of KMD values between the bounds defined and for the repeat unit defined between the parenthesis



Kendrick E., Anal. Chem. 1963
DOI: 10.1021/ac60206a048

3.3.6 Post treatment

- « Posttreatment » include multiple options that can be toggled on/off after the assignment process is finished
- Peaks that do not have any network connection will be removed (it will crash your assignment process if you do not use any network !)
- The three options below allow you to try and compute « chemical formulae » instead of ionic formulae. Let's try it, check the box before « Try to find adduct formula » and then press « Adducts list »
- The algorithm will try to remove a Na^+ ion for each assignment, if it is not possible, it will try with NH_4^+ , and if it is still not possible (for example if there is no nitrogen in the ionic formula), then it will try to remove H^+
- For the test dataset you can remove « Na^+ » by pressing DEL and « VALIDATE »



3.3.6 Posttreatment

- Let's then press « Customize output table columns ». This will allow you to modify the format of your results. You can use any column order, display additional data or even add the original columns of your peaklist, if you had for example the resolution of your peaks in a third column. Modify this as much as you need !
- For now, let's try to add our « chemical formula » from the previous slide. Press « Save and add a column »
- A column named « empty » will appear. Change it to « Chemical Formula » and then press « VALIDATE »
- Re-compute the analysis

Here is presented the output format table
Customize before attribution

Info	Special parameters for this data info			Delete
m/z Exp		▲	▼	DEL
Intensity	☒ Sum isotopes intensity	▲	▼	DEL
Ion Formula		▲	▼	DEL
m/z Calc		▲	▼	DEL
ppm		▲	▼	DEL
Ion Type	☐ write as boolean	▲	▼	DEL
DBE		▲	▼	DEL
All Atoms Cols.	Write to replace default els. list (O;C;...)	▲	▼	DEL
Elemental Ratios	H/C;O/C	▲	▼	DEL

Save and add a column

VALIDATE

Here is presented the output format table
Customize before attribution

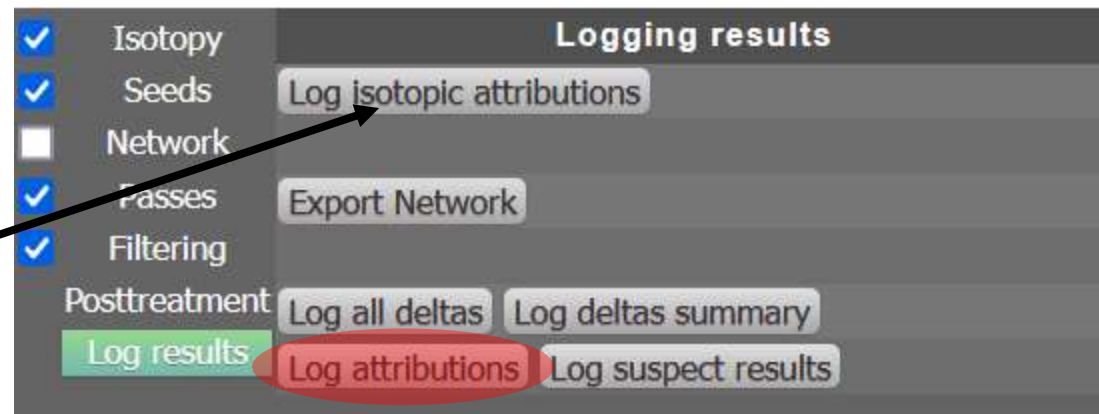
Info	Special parameters for this data info			Delete
m/z Exp		▲	▼	DEL
Intensity	☒ Sum isotopes intensity	▲	▼	DEL
Ion Formula		▲	▼	DEL
m/z Calc		▲	▼	DEL
ppm		▲	▼	DEL
Ion Type	☐ write as boolean	▲	▼	DEL
DBE		▲	▼	DEL
All Atoms Cols.	Write to replace default els. list (O;C;...)	▲	▼	DEL
Elemental Ratios	H/C;O/C	▲	▼	DEL
Chemical Formula	Please toggle on adduct search	▲	▼	DEL

Save and add a column

VALIDATE

3.3.6 Log results

- In this menu you will be able to look at the tagged isotopes



- You also have many log options for the network and the mass differences. This can be useful when investigating the assignment process of a sample or the most common mass differences
- Here, let's press « Log attributions ». The rightmost menu should be « chemical formula », with the adduct and the charge removed (the new column we just added)

	ionType	DBE	#C	#H	#O	#N	H/C	O/C	Chemical Formula
9482	adduct	3.5	6	7	3	0	1.1666666666666667	0.5	C6H6O3
977	adduct	3.5	6	7	4	0	1.1666666666666667	0.6666666666666666	C6H6O4
4215	adduct	2.5	6	9	4	0	1.5	0.6666666666666666	C6H8O4
1049	adduct	5.5	9	9	2	0	1	0.2222222222222222	C9H8O2
1726	adduct	5.5	8	7	3	0	0.875	0.375	C8H6O3
1119	adduct	4.5	9	11	2	0	1.2222222222222223	0.2222222222222222	C9H10O2
8728	adduct	4.5	8	9	3	0	1.125	0.375	C8H8O3
9456	adduct	3.5	7	9	4	0	1.2857142857142858	0.5714285714285714	C7H8O4
1613	adduct	2.5	6	9	5	0	1.5	0.8333333333333334	C6H8O5
3962	adduct	6.5	10	9	2	0	0.9	0.2	C10H8O2
7053	adduct	6.5	9	7	3	0	0.7777777777777778	0.3333333333333333	C9H6O3
18335	adduct	1.5	6	11	5	0	1.8333333333333333	0.8333333333333334	C6H10O5
1579	adduct	5.5	10	11	2	0	1.1	0.2	C10H10O2
3155	adduct	5.5	9	9	3	0	1	0.3333333333333333	C9H8O3
9824	adduct	4.5	9	11	3	0	1.2222222222222223	0.3333333333333333	C9H10O3

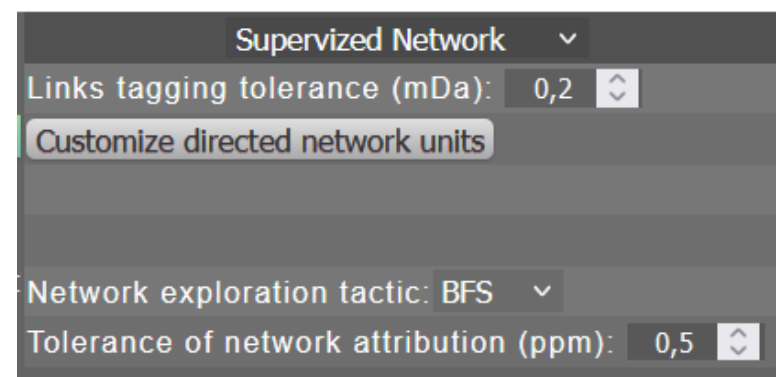
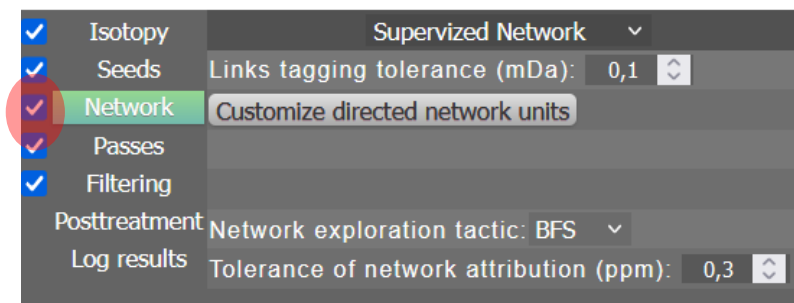
COPY TABLE

4. Network parameters

- There are two main network approaches. « Supervized network » where you select your own mass differences units, and « Unsupervised network » where the most common mass differences of your sample are searched and used to build the network.
- Choose « Supervized network » when you know the most common mass differences (CH_2 , H_2 ...) and « Unsupervised network » for any other case (part 4.2 of this tutorial)

4.1 Supervized network

- First, check the « Network » box to toggle on supervised network
- For the links tagging tolerance, use 0.2 mDa (as a first approach, the same tolerance used for the isotopy is good enough)
- Do not change the network exploration tactic unless Punc'data notifies you of network errors. If there are, then changing it could be useful to verify that taking different paths in the network does not change the assignments made.
- For the tolerance, we can put a slightly higher tolerance than the *De Novo* one. Let's put 0.5 ppm



4.1 Supervized network

- Press the button « customize directed network units »
- A menu opens to change the units. The mass is automatically computed.
- We are analyzing cellulose. Let's try to add the saccharide monomer, which is C₆H₁₀O₅.
- Press « SAVE AND ADD A MASS », and write C₆H₁₀O₅ in the « Formula » column
- Press « VALIDATE »

Edit the database here

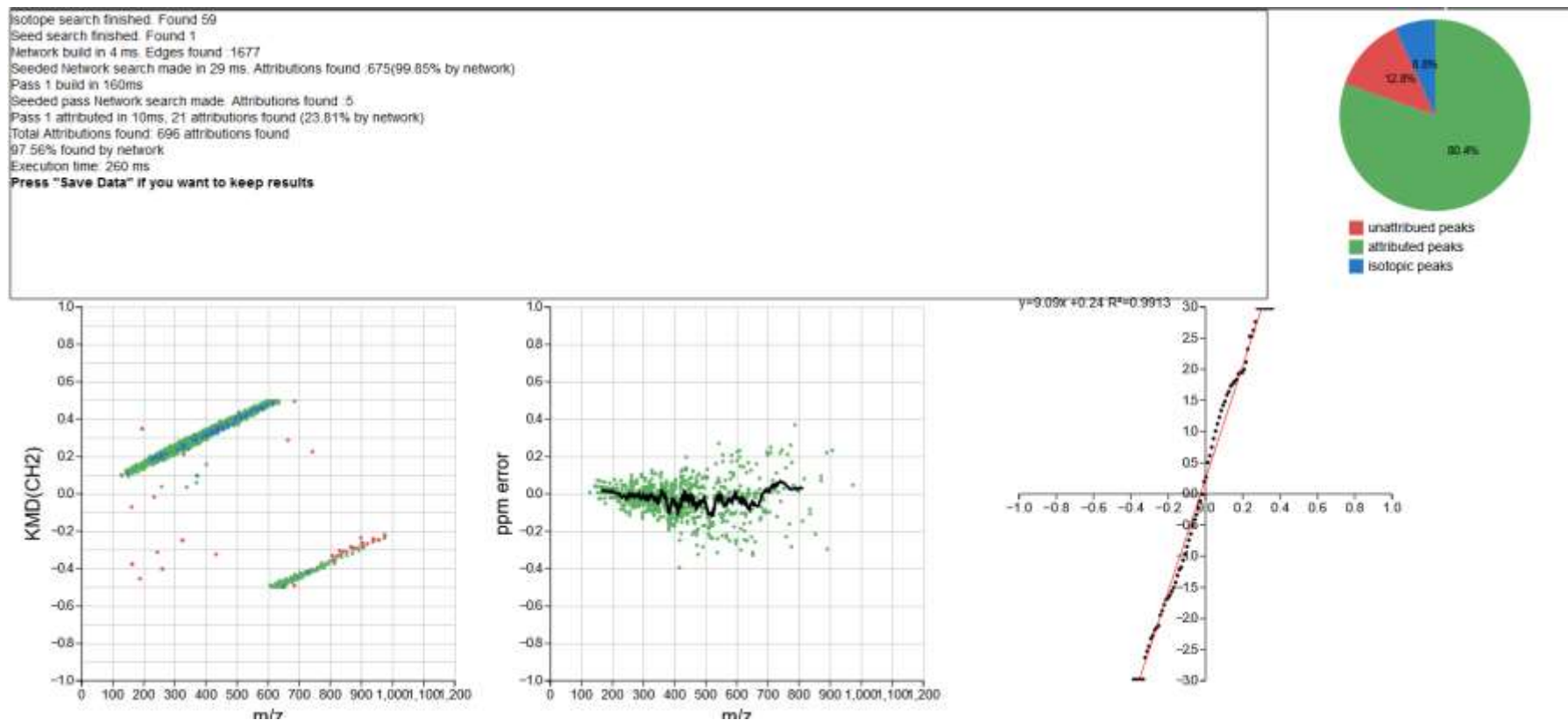
Formula	Mass	DEL
CH ₂	14,01565	DEL
H ₂	2,01565	DEL
H ₂ O	18,010565	DEL
C ₆ H ₁₀ O ₅	162,052825	DEL

Save and add a mass

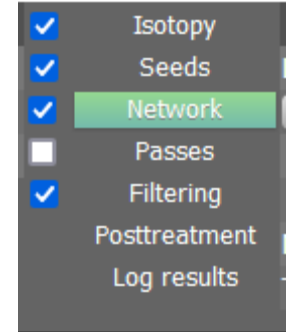
VALIDATE

4.1 Supervized network

- We customized our network. Press « COMPUTE ». New results should appear.



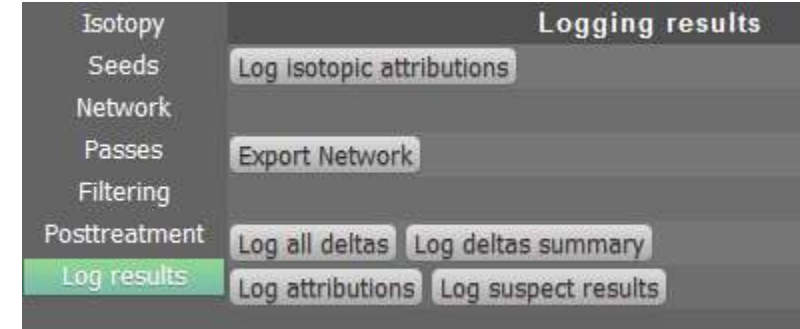
4.1 Supervized network



- We can now uncheck the « Passes » step, and then press « Compute »
- Slightly less assignments are found, but the algorithm is faster ! This can be quite noticeable on big datasets with >7000 peaks.
- You could still try and add more network units or more seeds to connect the lost assignments.

4.1 Supervized network

- If you want to look at your network results, press « Log results »
- « Log all deltas » will show you every mass difference found, with an indication of the two m/z values it connects
- « Log deltas summary » will display each mass difference type and the number of times it was found in the dataset



Data about the edges of the network:

Link type	# Occurences
CH2	420
H2	390
H2O	520
C6H10O5	347

COPY

4.2 Unsupervized network

- On the top line, change the value to « Unsupervized Network »
- « Deltas grouping tolerance » defines how the algorithm groups together mass differences under the same $\Delta m/z$ value. Let's set it to the same value as your isotopy tagging tolerance (in this case, 0.2 mDa)
- The bounds define the range of mass differences that are searched. If you know that your sample may contain a polymer with a repeat unit >100 Da, increase the max value. Usually the most common mass difference have low $\Delta m/z$ value. 1-100 is already good here.
- « Keep » indicates how many of the most common mass differences are kept to build the network. 10 is already more than enough in most cases.

Isotopy	Unsupervized Network ▾	
Seeds	Deltas grouping tolerance(mDa): 0,1 ▴ ▾	
Network	Bounds(Da): 1 ▴ ▾ - 100 ▴ ▾ Keep: 10 ▴ ▾	
Passes	Custom attrib. DB Custom attrib. pass	
Filtering	Delta attribution tolerance (mDa): 0,1 ▴ ▾	
Posttreatment	Network exploration tactic: BFS ▾	
Log results	Tolerance of network attribution (ppm): 0,5 ▴ ▾	

4.2 Unsupervized network

- Once the most common mass differences are found, Punc'data will try to assign them. It will look first through a database (customized through the button « custom attrib. DB ») and then through a small De Novo pass (customized through custom attrib. Pass »)

For the test dataset, you will not need to modify anything

TIP: You can use negative number of elements in the custom attrib. pass

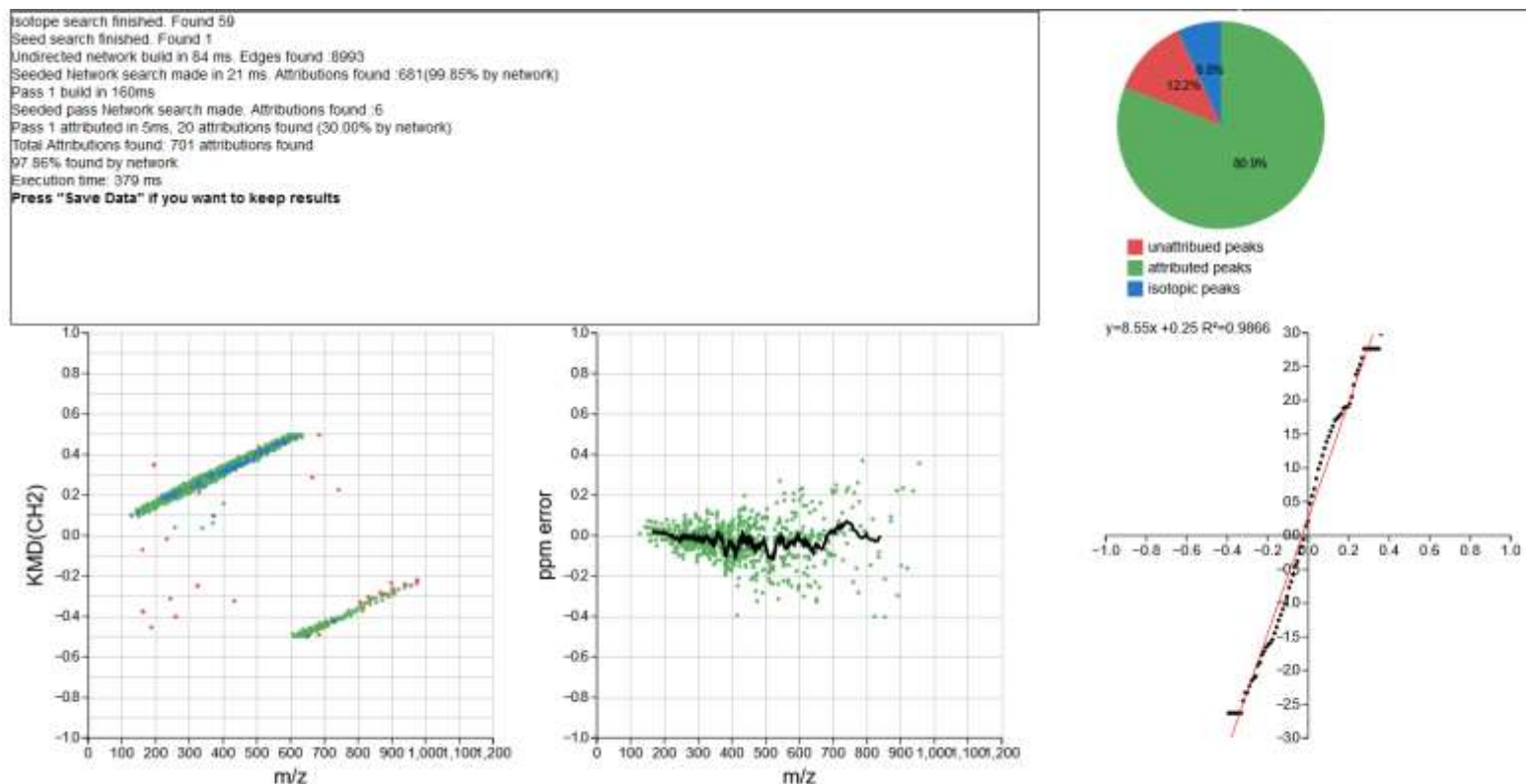
- The assignment tolerance of all mass differences is customized through the option « Delta attribution tolerance ». Keep it at 0.1 mDa.
- For the last two options, please refer to slide #27

Isotopy	Unsupervised Network ▾	
Seeds	Deltas grouping tolerance(mDa): 0,1 ▾	
Network	Bounds(Da): 1 ▾ - 100 ▾ Keep: 10 ▾	
Passes	Custom attrib. DB Custom attrib. pass	
Filtering	Delta attribution tolerance (mDa): 0,1 ▾	
Posttreatment	Network exploration tactic: BFS ▾	
Log results	Tolerance of network attribution (ppm): 0,5 ▾	

Green and Perdue, Anal. Chem. 2015
DOI: <https://doi.org/10.1021/ac504166t>

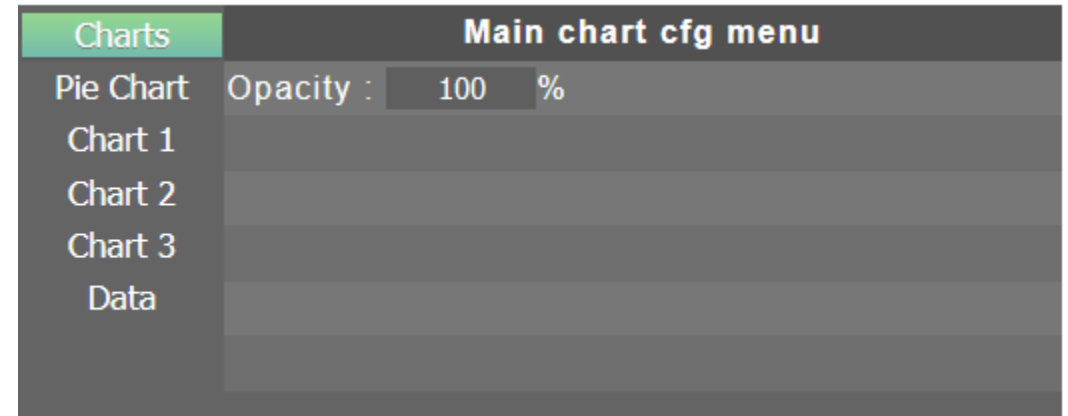
4.2 Unsupervised network

- Leave the parameters as is and press « Compute ». You should get more assignments



5. Customize assignment charts

- The charts on this tab can be customized with the middle-right menu
- « CHARTS » is currently only for dots opacity
- « PIE CHART » changes which sectors are displayed on the pie chart and if the sectors are relative to number of assignments or to total intensity
- « CHARTS 1/2/3 » can help you modify the three charts displayed below the text box
- « DATA » allows you to hide some subsets of data on all three charts



TIP: Many more chart type and functions/customisability exist in Punc'data, but are only available in « CANVAS » and « STATS »

5. Customize assignment charts

- Let's change Chart 1. Press « Chart 1 ». C2H4 is not adapted for cellulose. You can select another repeat unit in the menu on the 4th line, or press the button on the 5th line, write « CH₂O » and press enter.

- You can add a Kendrick divisor

Fouquet et al. Mass Spectrometry 2017

DOI : 10.5702/massspectrometry.A0055

Charts	Type of cell: Kendrick plot
Pie Chart	X: [0 1200] m/z
Chart 1	Y: [-1 1] Exp m/z
Chart 2	Size: 2
Chart 3	Polyethylene(C2H4)
Data	28,0313 / 1

Charts	Type of cell: Kendrick plot
Pie Chart	X: [0 1200] m/z
Chart 1	Y: [-1 1] Exp m/z
Chart 2	Size: 2
Chart 3	Polyethylene(C2H4)
Data	CH2O 30,010565 / 1

5. Customize assignment charts

- For Chart 3, let's change the normal probability plot to a histogram.

Press « Chart 3 », and on the top title line select « Errors histogram »

Put the Y values between 0 and 100, and « Number of bars » at 50. Press « COMPUTE » and you will now get your new chart.

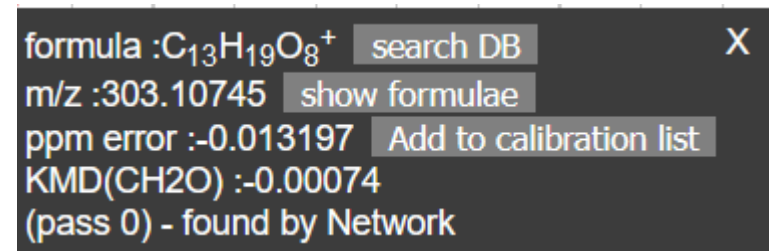
You can press « CAPTURE » on the top-right menu to save as SVG or PNG any chart.

Charts	Type of cell: Normal probability ▾
Pie Chart	X: [-1 ▾ 1 ▾]
Chart 1	Y: [-3 ▾ 3 ▾]
Chart 2	Sampling bins: 75 ▾
Chart 3	
Data	

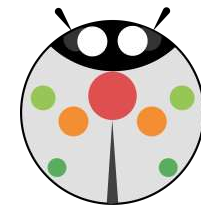
Charts	Type of cell: Errors histogram ▾
Pie Chart	X: [-1 ▾ 1 ▾]
Chart 1	Y: [0 ▾ 100 ▾]
Chart 2	Number of bars: 50 ▾
Chart 3	
Data	

5. Customize assignment charts

- Some Tips with charts:
 - You can hover over points to display additional data on them: m/z ratio, assigned formula, ppm error, the pass with which they were assigned (Pass 0 for seeds) and if they were assigned as a seed, through passes or through network.
 - You can select points to highlight them on other charts
 - Shift+drag allows you to zoom on charts
 - Ctrl+click makes a tooltip sticky. Additional buttons appear « Show formulae » shows you alternative formulae for this m/z ratio with a custom pass assignment.



formula :C₁₃H₁₉O₈⁺ [search DB](#) X
m/z :303.10745 [show formulae](#)
ppm error :-0.013197 [Add to calibration list](#)
KMD(CH₂O) :-0.00074
(pass 0) - found by Network



Do you have any questions ?

- Please adress any comments, suggestions or questions at :
- lcpa2mc-puncdata-contact@protonmail.com
- Or through GitHub : <https://github.com/WTVoe/puncdata/discussions>



MassLor
Spectrométrie de Masse